NETWORK MONITORING AND TROUBLESHOOTING TOOLS

Practical Network Administration Assignment

TASK 1

Install Wireshark on your laptop and capture two minutes of network traffic while browsing Instagram or YouTube. Save the capture as a .pcap file and note the top three protocols observed.

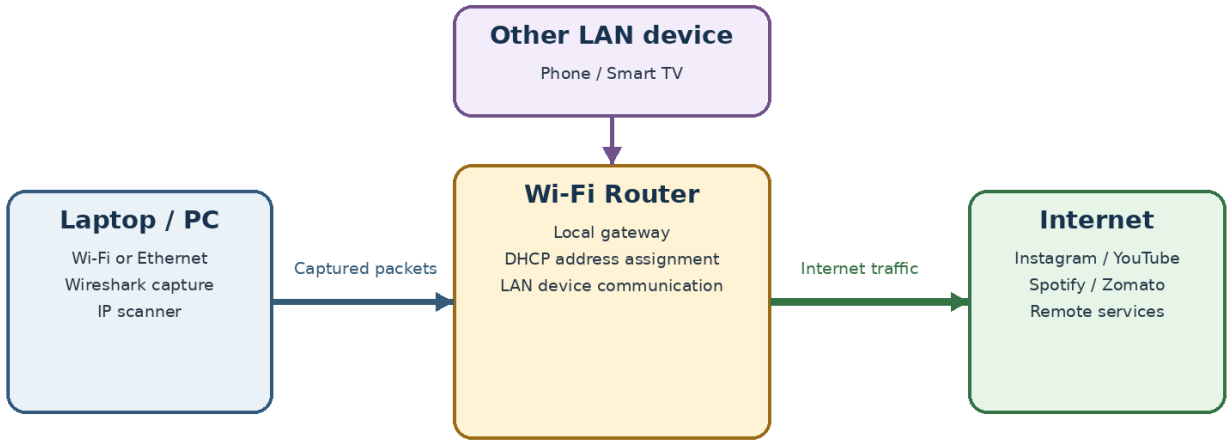
Answer

Wireshark is a network protocol analyser that captures packets from a selected network interface and displays details such as source and destination addresses, ports and protocol names. A capture can help identify the kinds of traffic generated by a browser, but encrypted application content (for example, HTTPS/TLS payloads) is not normally readable.

Procedure:

1. Download Wireshark from its official website (wireshark.org) and install it. On Windows, allow installation of Npcap if the installer offers it; Npcap enables packet capture.
2. Open Wireshark. Select the active Wi-Fi or Ethernet interface by checking which interface shows live activity.
3. Start the capture, open Instagram or YouTube in a browser, and browse normally for approximately two minutes. Avoid capturing other people's traffic; capture only on a network and device you are authorised to monitor.
4. After two minutes, select Stop. Use the Protocol column or Statistics → Protocol Hierarchy to identify the protocols present and their packet counts.
5. Save the capture using File → Save As. Choose the .pcap format if available, or .pcapng if that is the default format required by the installed Wireshark version. Use a clear filename such as Browser_Traffic_2_Minutes.pcap.

Network capture overview:



Wireshark records packets visible to the selected network interface.

A local IP scanner discovers responding devices only on a network you own or are authorised to test.

Common protocols you may see include DNS (domain-name lookups), TCP (reliable transport), TLS (encrypted sessions) and QUIC (often used by modern web services over UDP). The actual top three depend on the websites, browser, network and capture duration; do not report these examples as measured results.

Record your actual capture results:

Item	Your observation
Capture file name	_____
Network interface	_____
Capture duration	2 minutes (record actual start/end time: _____)
Top protocol 1 and packet count	_____
Top protocol 2 and packet count	_____
Top protocol 3 and packet count	_____
File saved successfully?	Yes / No

Submission evidence: Attach a screenshot of Wireshark's Protocol Hierarchy or packet list and submit the saved capture file. This document does not contain a fabricated capture; the protocol ranking must be filled in from your own .pcap file.

TASK 2

Use the ping command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com. Record response times and identify which site responds faster.

Answer

The ping utility sends Internet Control Message Protocol (ICMP) Echo Request packets and measures the time taken for replies. Some websites or networks block or deprioritise ICMP, so a timeout does not necessarily mean that the website itself is unavailable.

Windows procedure:

6. Open Command Prompt.
7. Run: ping -n 4 www.spotify.com
8. Record the time values for the replies and the final Average value, if replies are received.
9. Run: ping -n 4 www.zomato.com and record the same information.
10. Compare the average response times. The lower average indicates the faster ICMP response during that test, not necessarily the faster page-loading experience.

macOS / Linux procedure: open Terminal and run `ping -c 4 www.spotify.com`, then `ping -c 4 www.zomato.com`. Record the average or mean round-trip time shown by the terminal.

Measure	www.spotify.com	www.zomato.com
Packets sent	_____	_____
Packets received	_____	_____
Packet loss	_____ %	_____ %
Average response time	_____ ms	_____ ms
Result	Responding / No reply	Responding / No reply

Comparison:

The site with the smaller measured average response time is the faster responder for this particular test. If one or both hosts time out, write “ICMP reply unavailable” rather than assuming that the site is down; repeat the test later or use an approved application-level connectivity test.

Result to complete: Based on my measurements, _____ responded faster. Its average time was _____ ms compared with _____ ms for the other site.

TASK 3

Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop or any browser-based app you use. Record the process name and memory usage for each.

Answer

Operating systems often run several processes for one browser or desktop application. For example, Chrome may use separate processes for tabs, extensions, GPU rendering and background services. Memory usage changes over time and depends on the open tabs, extensions and workload.

Windows:

11. Press Ctrl + Shift + Esc to open Task Manager.
12. Select Processes. Expand Google Chrome or the relevant application group if it is shown.
13. Read the Memory column for each related process. If more detail is required, use the Details tab and look for process names such as chrome.exe or WhatsApp.exe.
14. Record each distinct relevant process and its memory usage in the table below. Do not add unrelated system processes.

macOS:

15. Open Activity Monitor from Applications → Utilities.
16. Select the CPU or Memory tab and use the search field to find Chrome, WhatsApp or the browser-based app.
17. Record the process name and Memory value for every relevant running process. If several rows share the same app name, list each row separately.

Application / group	Process name	Memory usage	Notes (tab/helper/GPU etc.)
Chrome	_____	_____ MB	_____
Chrome	_____	_____ MB	_____
WhatsApp Desktop	_____	_____ MB	_____
Other browser app	_____	_____ MB	_____
Other browser app	_____	_____ MB	_____

Comparison and observation:

Compare the recorded memory values to identify which process uses the most RAM. A browser may show many processes, so the largest individual process is not always the same as the total memory used by the complete application group. Record the time of observation because process memory changes continuously.

Evidence: Attach a screenshot of Task Manager or Activity Monitor with the relevant process names and Memory column visible. Do not close applications or terminate processes merely to make the screenshot look simpler.

TASK 4

Download and run the free version of Angry IP Scanner or Advanced IP Scanner to scan your local Wi-Fi network. List the IP address and device name of at least two devices connected to the network.

Answer

A local IP scanner checks a specified address range and reports devices that respond to the scanner's discovery methods. Device names may be missing when a device does not provide a hostname, blocks discovery, or is asleep. Only scan a home or college network that you own or have explicit permission to test.

Procedure:

18. Download either Angry IP Scanner from its official project website or Advanced IP Scanner from its official website. Install and launch the selected tool.
19. Find your computer's local IPv4 address and subnet mask. On Windows, run `ipconfig` in Command Prompt; on macOS, check Network settings or use an approved network command.
20. Determine the local subnet. For example, an address of 192.168.1.25 with subnet mask 255.255.255.0 commonly corresponds to 192.168.1.1–192.168.1.254. Use your actual subnet, not this example.
21. Enter the correct local IP range in the scanner and start the scan. Do not scan public IP ranges or networks outside your authorisation.
22. Review the results for responding devices. Record at least two IP addresses and their displayed hostnames/device names. If a hostname is blank, write “Not resolved” rather than inventing a name.

Network discovery context:

The Wi-Fi router is normally the gateway for local devices. A scanner running on a laptop sends discovery probes within the local subnet and lists devices that respond. The network diagram in Task 1 shows the laptop, router and other devices on this local network.

No.	IP address	Device name / hostname	Status / notes
1	_____	_____	Responded / other
2	_____	_____	Responded / other
3 (optional)	_____	_____	Responded / other

Safety note: Do not disable your firewall globally just to make a scan work. If discovery is blocked, follow your institution's or network administrator's approved procedure and scan only the authorised subnet.

Evidence: Attach a screenshot showing the scanned address range and at least two discovered devices. Hide or redact any device details that you are not permitted to share.

TASK 5

Use ChatGPT or Bing AI to suggest three hardware tools used in network troubleshooting, such as a cable tester, crimping tool or multimeter. For each, write one line describing when you would use it.

Answer

Hardware tools help a technician locate physical-layer faults, verify cabling and check electrical conditions. Three useful tools are listed below.

Hardware tool	Main purpose	When would you use it?
Ethernet cable tester	Checks continuity and correct pin-to-pin wiring of a network cable.	Use it when a PC has no link or an Ethernet cable may have an open wire, short or incorrectly ordered pairs.
RJ45 crimping tool	Attaches an RJ45 modular plug to the end of a twisted-pair Ethernet cable.	Use it when making or repairing a cable end after cutting away a damaged connector.
Digital multimeter	Measures electrical quantities such as voltage, resistance and continuity.	Use it when checking a power adapter, verifying continuity in suitable low-voltage circuits, or investigating a power-related fault.

Comparison of the tools:

Tool	Type of problem it helps diagnose	Typical result
Cable tester	Cable wiring and continuity faults	Pass/fail or wire-map indication
RJ45 crimping tool	Poor or damaged cable termination	A correctly terminated connector when used properly
Digital multimeter	Power and electrical continuity faults	Measured voltage, resistance or continuity

Additional note: A crimping tool is mainly used to create or repair a cable termination; it does not by itself certify that the finished cable meets a network performance standard. A cable tester should be used after termination to verify the wire map.

Conclusion

Wireshark helps analyse packet-level network activity, ping provides a basic latency and reachability check, Task Manager or Activity Monitor reveals application resource use, and IP scanners help inventory authorised local devices. Physical tools such as cable testers, crimping tools and multimeters complement software tools by helping identify cabling and power faults.